

DONNA NOOK, United Kingdom

WMO: 033850

Lat: 53.475N

Lon: 0.153E

Elev: 8

StdP: 101.23

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.9	-0.7	-4.9	2.5	0.3	-3.7	2.8	1.0	17.9	5.3	16.0	5.6	4.5	290	0.724

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.1	23.4	17.9	22.0	17.1	20.7	16.6	19.0	21.9	18.1	20.7	17.3	19.7	6.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	12.8	20.2	17.0	12.2	19.5	16.2	11.6	18.6	53.9	21.9	51.2	20.8	48.8	19.9	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.2	13.6	12.1	DB	-4.7	27.6	2.4	2.0	-6.5	29.0	-7.9	30.2	-9.3	31.3	-11.0	32.7	(4)
			WB	-4.9	21.3	2.1	1.9	-6.4	22.7	-7.7	23.8	-8.9	24.8	-10.4	26.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	4.7	5.1	6.4	8.4	11.0	13.8	16.0	16.2	14.1	11.3	7.5	5.1
	DBStd	4.91	2.59	2.90	2.76	2.63	2.48	2.40	2.21	2.16	2.20	2.66	2.75	3.13
	HDD10.0	756	166	138	117	63	17	1	0	0	1	17	83	154
	HDD18.3	3059	423	370	371	299	228	137	77	72	127	219	326	411
	CDD10.0	757	1	2	4	13	47	116	187	194	125	57	8	2
	CDD18.3	16	0	0	0	0	0	2	6	7	1	1	0	0
	CDH23.3	50	0	0	0	0	1	9	20	16	4	1	0	0
	CDH26.7	4	0	0	0	0	0	1	1	2	0	0	0	0
Wind	WSAvg	6.0	6.8	6.9	6.3	6.0	5.8	5.2	4.9	5.2	5.6	6.3	6.4	6.7
Precipitation	PrecAvg	641	57	43	37	43	49	56	59	60	51	66	63	56
	PrecMax	818	103	72	73	125	116	175	145	109	120	119	120	113
	PrecMin	435	13	12	2	2	16	10	20	6	9	22	21	21
	PrecStd	108	26	15	18	32	25	38	32	30	29	24	24	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	14.1	15.8	19.1	21.0	24.5	25.9	25.1	23.5	20.4	15.7	13.6
		MCWB	11.1	11.6	11.9	13.9	15.9	18.3	18.9	18.8	17.9	16.6	13.8	11.8
	2%	DB	11.3	12.3	13.7	16.2	18.7	21.7	23.3	23.0	21.2	17.8	13.9	12.4
		MCWB	9.7	10.1	10.6	12.1	14.3	17.0	17.9	17.8	17.0	14.9	12.3	11.0
	5%	DB	10.0	10.8	12.2	14.4	16.9	19.9	21.7	21.7	19.3	16.2	12.7	11.1
		MCWB	8.6	9.0	9.6	11.0	13.1	16.0	17.2	16.9	15.8	14.1	11.2	9.8
	10%	DB	8.8	9.6	10.9	12.8	15.4	18.3	20.3	20.4	17.9	15.1	11.5	9.7
		MCWB	7.5	8.0	8.6	10.0	12.1	14.8	16.5	16.4	14.9	13.3	10.3	8.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	12.0	12.5	14.8	16.6	19.4	20.1	20.1	19.0	17.0	14.1	12.1
		MCDB	12.7	13.7	14.7	17.6	19.8	22.9	23.9	23.0	21.6	20.1	15.3	13.2
	2%	WB	9.9	10.4	11.1	12.9	14.9	17.6	18.9	18.9	17.6	15.6	12.6	11.1
		MCDB	11.1	12.0	13.1	15.3	17.8	20.8	22.0	21.5	19.9	17.2	13.7	12.3
	5%	WB	8.8	9.2	9.9	11.4	13.7	16.4	18.0	18.0	16.5	14.5	11.5	9.9
		MCDB	9.8	10.6	11.8	13.7	16.2	19.1	20.5	20.5	18.4	15.7	12.4	10.9
	10%	WB	7.6	8.1	8.9	10.3	12.6	15.4	17.2	17.3	15.6	13.6	10.5	8.7
		MCDB	8.6	9.4	10.6	12.3	14.8	17.7	19.5	19.4	17.2	14.8	11.3	9.6
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.2	6.0	6.4	6.3	6.7	7.0	7.1	6.6	5.7	5.1	4.6
		MCDBR	6.0	6.6	8.1	9.5	9.5	9.8	9.6	9.1	8.9	6.9	5.7	5.8
		MCWBR	5.5	5.2	5.7	6.1	5.8	5.9	5.4	5.0	5.5	5.0	4.8	5.2
	5% WB	MCDBR	5.9	6.5	7.8	8.8	8.5	9.1	8.4	7.6	7.7	6.5	5.5	5.8
		MCWBR	5.6	5.4	5.9	6.2	5.7	6.1	5.3	4.9	5.4	5.2	4.9	5.3
Clear-Sky Solar Irradiance	taub	0.328	0.344	0.375	0.402	0.400	0.407	0.409	0.398	0.377	0.357	0.329	0.316	
	taud	2.347	2.385	2.302	2.251	2.313	2.313	2.320	2.371	2.425	2.448	2.402	2.329	
	Ebn at Noon	635	754	800	826	849	844	834	823	795	729	632	571	
	Edh at Noon	59	77	102	121	120	121	119	107	90	72	56	50	
All-Sky Solar Radiation	RadAvg	0.65	1.34	2.53	3.98	5.02	5.18	5.02	4.23	3.05	1.71	0.81	0.51	
	RadStd	0.06	0.18	0.24	0.41	0.37	0.39	0.44	0.28	0.26	0.13	0.08	0.05	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page